# COMP2300 Set 7.5 Practice Exam

Topics: Locality, Cache Basics, Associativity, Write Policies, SRAM, DRAM

**Time:** 120 minutes **Total:** 100 marks

**Instructions**

- Answer in English.
- Part A is multiple choice.
- Part B contains cache calculations and memory-technology reasoning.
- Part C contains adapted past-exam style questions with sources.

## Part A: Multiple Choice (30 marks, 3 marks each)

Choose exactly one option for each question.

**A1.** Temporal locality means:

(A) Nearby addresses are likely to be used
(B) Recently used data is likely to be used again soon
(C) Memory always grows upward
(D) Every miss is compulsory

**A2.** A cache is best described as:

(A) A large slow memory for archival storage
(B) A small fast storage layer that keeps a subset of a larger slower memory
(C) A branch predictor
(D) A decode-stage register

**A3.** In a direct-mapped cache, each set contains:

(A) 0 lines
(B) 1 line
(C) 2 lines
(D) 4 lines

**A4.** Which policy updates lower memory immediately on a write hit?

(A) Write-back
(B) Write-through

(C) No-write-allocate

(D) Random replacement

**A5.** Which memory technology typically backs on-chip caches?

(A) DRAM

(B) SRAM

(C) Magnetic tape

(D) SSD flash only

**A6.** Which statement about DRAM is true?

(A) It stores bits in cross-coupled inverters

(B) It never needs refresh

(C) Its read operation is destructive

(D) It is always faster than SRAM

**A7.** Which component selects a row in a memory array?

(A) Decoder / wordline logic

(B) BTB

(C) ALU

(D) Link register

**A8.** A 2-way set associative cache allows each block to be placed in:

(A) Any line in the whole cache

(B) Exactly one of two lines in the indexed set

(C) Exactly one line in the whole cache

(D) Main memory only

**A9.** Which storage element is the most expensive per bit but fastest among the three?

(A) DRAM

(B) SRAM

(C) Flip-flops

(D) Magnetic disk

**A10.** In a memory hierarchy, the usual trend is:

(A) Larger memories are always faster

(B) Smaller memories tend to be faster

(C) DRAM is faster than registers

(D) Cache eliminates the need for main memory

## Part B: Computational Problems (50 marks)

**B1. Hit and miss rates (8 marks)**

A cache sees 500 memory accesses, of which 40 are misses.

(a) Compute the miss rate.

(b) Compute the hit rate.

**B2. Address breakdown (8 marks)**

A direct-mapped cache has 4 sets and a block size of 8 bytes.

(a) How many block-offset bits are required?

(b) How many set-index bits are required?

(c) In a 16-bit address, how many tag bits remain?

**B3. Direct-mapped placement (8 marks)**

Consider the cache from B2. For address `0x0034`:

(a) Give the binary form of the low 5 bits.

(b) Identify the block offset.

(c) Identify the set index.

**B4. Cache benefit calculation (8 marks)**

A device has the following access costs:

- Cache hit: 1 cycle
- Cache miss: 4 cycles
- Cache disabled: 3 cycles

(a) Compute the average access time when hit rate is 25%.

(b) Compute the average access time when hit rate is 50%.

(c) For each case, state whether enabling the cache is beneficial.

**B5. SRAM vs DRAM reasoning (8 marks)**

Explain, in one or two short sentences each:

(a) Why SRAM is preferred for caches.

(b) Why DRAM is preferred for large main memory.

**B6. Write-policy comparison (10 marks)**

Briefly compare the following pairs:

(a) write-through vs write-back

(b) write-allocate vs no-write-allocate

For each pair, give one advantage and one tradeoff.

## Part C: Past Exam Questions (Source-Tagged) (20 marks)

**C1. Cache enable decision (Source: exam2025-main.pdf, Q1, adapted) (8 marks)**
A device access costs 3 cycles with cache disabled. With cache enabled, hits cost 1 cycle and misses cost 4 cycles. Derive the condition on hit rate under which the cache is beneficial.

**C2. Levels in the memory hierarchy (Source: 2023_Final_Exam_SuppDA.pdf, microarchitecture MCQ topic, adapted) (6 marks)**
List a reasonable top-to-bottom memory hierarchy for a modern processor and explain why each lower level tends to be larger but slower.

**C3. Cache misses in pipelined execution (Source: exam24r.pdf / exam24px1.pdf, Q7c context, adapted) (6 marks)**
Explain why long-latency cache misses make memory-level parallelism valuable in later out-of-order designs, even though the basic cache concepts are introduced earlier.